

Phishing Email Detection Model

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~/phishing-email-detector/e-mail.py - Mousepad

1 import pandas as pd
2 from sklearn.model_selection import train_test_split
3 from sklearn.feature_extraction.text import CountVectorizer
4 from sklearn.naive_bayes import MultinomialNB
5 from sklearn.metrics import accuracy_score, classification_report
6
7 data = {
8     'text': [
9         "Congratulation! You've won a lottery. Click here to claim.",
10        "Your invoice from our recent purchase is attached.",
11        "Urgent: Verify your account immediately!",
12        "Hello! Just checking in to see how you're doing.",
13        "Important: update your payment information.",
14        "Meet your new colleague at the office."
15    ],
16    'label': [1,0,1,0,1,0]
17 }
18
19 df = pd.DataFrame(data)
20
21 print('Dataset: ')
22 print(df)
23
24 X = df['text']
25 y = df['label']
26
27 X_train, X_test, y_train, y_test = train_test_split(X,y,test_size=0.2, random_state=42)
28 vectorizer = CountVectorizer()
29 X_train_vectorized = vectorizer.fit_transform(X_train)
30 X_test_vectorized = vectorizer.transform(X_test)
31
32 model = MultinomialNB()
33 model.fit(X_train_vectorized,y_train)
34
35 y_pred = model.predict(X_test_vectorized)
36
37 accuracy = accuracy_score(y_test, y_pred)
38 report = classification_report(y_test, y_pred)
39
40 print(f'\n Accuracy: {accuracy:.2f}')
41 print("Classification Report:")
42 print(report)
43
```

```
(venv)blackking@kali: ~/phishing-email-detector~
Session Actions Edit View Help

(venv)~(blackking@kali)-[~/phishing-email-detector~]
$ python -c "import sklearn; print(sklearn.__version__)"
1.9.1

(venv)~(blackking@kali)-[~/phishing-email-detector~]
$ cd ~/phishing-email-detector~
source venv/bin/activate
python e-mail.py
Dataset:
      text  label
0  Congratulation! You've won a lottery. Click he...      1
1  Your invoice from our recent purchase is attac...      0
2      Urgent: Verify your account immediately!      1
3  Hello! Just checking in to see how you're doing.      0
4      Important: update your payment information.      1
5      Meet your new colleague at the office.      0

Accuracy:  0.00
Classification Report:
      precision    recall  f1-score   support

      0       0.00       0.00       0.00        1.0
      1       0.00       0.00       0.00        1.0

 accuracy          0.00
 macro avg         0.00       0.00       0.00        2.0
weighted avg         0.00       0.00       0.00        2.0

(venv)~(blackking@kali)-[~/phishing-email-detector~]
$
```